FORM 2

THE PATENTS ACT, 1970

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The Patents Rules, 2003

COMPLETE SPECIFICATION

(see section 10)

AI-Based Predictive Thermal Monitoring and Adaptive Charging Optimization System for Preventing Battery Swelling in Consumer Smartphones and Laptops

The following specification particularly describes the invention and the manner in which it is to be performed.

FIELD OF INVENTION

The present invention relates generally to the field of battery management, electronic device safety, and Edge Artificial Intelligence (AI). More particularly, the invention relates to a software-based, on-device predictive battery safety framework for smartphones, laptops, and other battery-powered electronic devices, configured to predict battery swelling risk in real time based on thermal, electrical, battery-health, application-level, processor workload, and user charging-behavior parameters.

BACKGROUND OF THE INVENTION

The widespread adoption of lithium-ion batteries in portable electronic devices, including smartphones, laptops, tablets, and other battery-powered consumer devices, has resulted in increasing concerns regarding battery swelling and associated safety risks. Battery swelling may occur due to thermal stress, overcharging, battery aging, prolonged high-current operation, and unmanaged device usage patterns. During excessive heating, particularly during charging or intensive device operation, accelerated electrochemical reactions may occur within the battery, resulting in the generation and accumulation of gases within the battery cell. The resulting internal pressure may cause physical expansion of the battery, which can lead to screen displacement, enclosure deformation, degradation of battery performance, reduced operational lifespan, and, in severe circumstances, thermal runaway, fire, or chemical leakage.

Conventional battery management and thermal protection mechanisms used in consumer electronic devices are predominantly reactive and threshold-based. Such mechanisms generally monitor parameters such as battery temperature, charging current, voltage, or state of charge and initiate protective actions only when predetermined threshold values are reached or exceeded. Consequently, corrective measures may be initiated after substantial thermal stress or electrochemical degradation has already occurred. Moreover, fixed thresholds may not adequately account for differences in battery age, historical charging behavior, application workload, ambient conditions, or individual device usage patterns.

Existing battery monitoring approaches also provide limited capability for predictive swelling-risk assessment. In particular, conventional systems generally lack mechanisms for continuously learning user-specific charging and application-usage behavior, identifying thermal patterns

associated with increased swelling risk, dynamically optimizing charging conditions, and recalibrating safety thresholds according to battery aging and changing operating conditions. As a result, the same protection parameters may be applied to batteries having substantially different health conditions and usage histories.

A US patent US11079212B2 titled “Circuitry and techniques for determining swelling of a battery/cell and adaptive charging circuitry and techniques based thereon” describes a technique for estimating battery swelling based on battery capacity fade/state of health, sensing swelling, and managing battery swelling through adaptive charging.

One more US patent US20260110744A1 titled “Detect and prevent battery swelling” describes a system that monitors battery charging data over time, including temperature, charge level, charging duration, battery age, cycle count, historical temperature and charging behavior, and uses accumulated usage conditions to identify circumstances that may lead to future swelling.

Further, although certain artificial-intelligence-assisted battery prediction systems have been proposed, such systems may rely on cloud-based data processing and model inference. Such dependence can introduce communication latency, require continuous network connectivity, increase exposure of device-usage and battery-related data to external processing environments, and reduce the effectiveness of predictive protection during offline operation. Cloud-dependent approaches may therefore be unsuitable for real-time, privacy-preserving safety intervention directly at the device level.

Therefore, there remains a need for an improved battery safety solution capable of predicting battery swelling risk before critical thermal conditions are reached, while continuously analyzing battery and device operating parameters.

OBJECTIVES OF THE INVENTION

The principal objective of the present invention is to provide a fully on-device Edge Artificial Intelligence (AI) framework configured to predict battery swelling probability in real time for smartphones, laptops, and other battery-powered electronic devices, without requiring cloud connectivity, internet access, or additional external hardware.

Another objective of the present invention is to dynamically optimize charging current, charging rate, and charging conditions based on continuously determined Thermal Stress Scores (TSS) and Swelling Probability Scores (SPS), thereby enabling proactive mitigation of thermal degradation and battery swelling before critical operating conditions arise.

One more objective is to provide a Behavioral Charging Learning Module configured to continuously analyze user charging habits, device usage patterns, charging duration, charging frequency, and related operational behavior to provide personalized predictive thermal protection and adaptive charging optimization.

Yet another objective is to provide an Application-Level Heat Mapping Module configured to identify and quantify thermal contributions associated with individual applications and workloads, and to incorporate the identified thermal contribution into processor-aware charging orchestration.

Significant objective is to provide a CPU/GPU-Aware Charging Control Module configured to coordinate charging parameters with processor workload conditions, thereby reducing concurrent heat generation and minimizing thermal stress during high computational activity.

Another objective is to provide a Battery Age Compensation Module configured to dynamically recalibrate thermal-risk thresholds, swelling-risk parameters, and charging intervention conditions according to battery aging indicators, charging-cycle history, degradation characteristics, and long-term battery health.

Further objective is to provide a Micro Thermal Gradient Analysis Module configured to detect localized temperature variations, abnormal thermal gradients, and early-stage thermal instability, thereby enabling intervention before substantial temperature escalation occurs.

Another objective is to continuously generate the Thermal Stress Score (TSS) and Swelling Probability Score (SPS) using multi-parameter thermal, electrical, behavioral, workload, and battery-health information for real-time battery safety decision-making.

Another objective is to provide a Self-Learning Feedback Engine employing lightweight on-device learning, including reinforcement-learning-based adaptation, to continuously improve swelling-risk prediction, charging optimization, thermal protection, and personalization based on accumulated operational feedback.

A further objective is to provide a fully autonomous software-only battery safety orchestration framework capable of operating completely offline while improving battery lifespan, reducing thermal degradation, enhancing user safety, and eliminating dependence on cloud-based processing or external sensing hardware.

SUMMARY OF THE INVENTION

The following summary is provided to facilitate a clear understanding of the new features in the disclosed embodiment and it is not intended to be a full, detailed description. A detailed description of all the aspects of the disclosed invention can be understood by reviewing the full specification, the drawing and the claims and the abstract, as a whole.

The present invention provides a fully on-device Edge Artificial Intelligence (AI)-based battery safety framework configured to continuously monitor and process battery sensor data, thermal parameters, charging conditions, device usage patterns, application-level workloads, processor activity, and battery aging characteristics for real-time prediction and prevention of battery swelling in smartphones, laptops, and other portable electronic devices.

In an aspect, the framework is configured to determine a Thermal Stress Score (TSS) and a Swelling Probability Score (SPS) based on the monitored parameters and to predict a potential swelling condition before occurrence of critical thermal or electrochemical conditions.

The framework dynamically optimizes charging current, charging rate, and charging conditions based on the determined TSS, SPS, battery health, processor workload, and operating conditions, thereby reducing thermal stress and preventing or mitigating battery degradation and swelling.

The framework further incorporates behavioral charging learning for identifying user-specific charging habits and device usage patterns and utilizes the learned behavior for personalized charging and thermal protection.

The invention further provides application-level heat mapping for determining thermal contributions of individual applications and CPU/GPU-aware charging control for coordinating charging operation with processor workload.

A battery age compensation mechanism dynamically recalibrates thermal-risk thresholds and charging intervention parameters based on battery aging and degradation characteristics. A micro thermal gradient analysis mechanism detects localized thermal abnormalities and early-stage thermal instability that may precede critical temperature escalation.

The framework further employs a self-learning feedback engine using lightweight on-device learning to continuously improve swelling prediction, charging optimization, and behavioral adaptation based on accumulated operational feedback.

Accordingly, the present invention provides an autonomous, software-only, privacy-preserving and offline-capable battery safety solution that enables predictive swelling prevention, adaptive thermal management, optimized charging, improved battery lifespan, and enhanced user safety without requiring cloud processing, internet connectivity, or additional external hardware.

BRIEF DESCRIPTION OF THE DRAWINGS

The manner in which the present invention is formulated is given a more particular description below, briefly summarized above, may be had by reference to the components, some of which is illustrated in the appended drawing It is to be noted; however, that the appended drawing illustrates only typical embodiments of this invention and are therefore should not be considered limiting of its scope, for the system may admit to other equally effective embodiments.

Throughout the drawings, the same drawing reference numerals will be understood to refer to the same elements and features.

The features and advantages of the present invention will become more apparent from the following detailed description a long with the accompanying figures, which forms a part of this application and in which:

Fig. 1 illustrates the Schematic for the room-temperature one-pot in-situ synthesis of Pt NPs/rGO-based nanocomposite hydrogel;

Fig. 2 illustrates the pXRD pattern of different nanocomposite formulations;

Fig. 3 illustrates the Process schematic describing the preparation of Pt NPs/rGO/Alg-Hy hydrogel;

Fig. 4 illustrates the (a) XRD patterns and (b) FTIR spectra for Pt NPs/rGO/Alg-Hy hydrogel sample and corresponding control samples;

Fig. 5 illustrates the Raman spectroscopic analysis of GO and Pt NPs/rGO/Alg-Hy hydrogel;

y showing stability over 16 h in b) acidic, d) alkaline, and f) neutral media;

REFERENCE NUMERALS

101 - Data Acquisition Layer

102 - Edge AI Analytics Layer

103 - Micro Thermal Gradient Analysis Layer

104 - Behavioral Intelligence & App Thermal Analysis Layer

105 - Adaptive Charging Orchestration Layer

105a – Reduce Charging Current

105b – Pause/Resume Charging

105c – CPU/GPU Workload Control

106 - Reduce Charging Current

107 - Pause / Resume Charging

108 - CPU/GPU Workload Control

109 - Self-Learning Optimization Layer

110 - Battery Safety Response Layer

DETAILED DESCRIPTION OF THE INVENTION

The present invention will now be described in detail with reference to the accompanying drawing. The following description is provided to illustrate the invention and should not be construed as limiting the scope of the invention.

The present invention operates on an Edge AI-driven predictive thermal monitoring and adaptive charging orchestration framework designed specifically for consumer smartphones, laptops, and portable electronic devices. The system intelligently integrates real-time battery sensor acquisition (101), swelling probability prediction (102), behavioral charging learning (104), application-level thermal analysis (104), CPU/GPU-aware charging optimization (105), battery age compensation (105), and self-learning reinforcement optimization (109) in order to prevent battery swelling before critical thermal conditions occur.

Unlike conventional battery management systems that primarily rely on fixed threshold-based reactive protection mechanisms, the proposed framework continuously forecasts thermal instability and swelling probability in advance and autonomously intervenes before irreversible electrochemical degradation develops.

Initially, the system continuously acquires operational data from battery and device sensors already embedded within modern smartphones and laptops through native operating system interfaces such as the BatteryManager API in Android devices, the IOKit framework in macOS and iOS devices, and the ACPI power management interface in Windows laptops. The acquired parameters include battery surface temperature, terminal voltage, charging and discharging current, State of Charge (SoC), State of Health (SoH), cumulative charge cycle count, CPU utilization, GPU workload, active application identifiers, application-level processor consumption metrics, charging duration, and ambient temperature where available. The collected data is timestamped and stored within a continuously maintained rolling time-series buffer for real-time predictive analysis (101).

The collected sensor information is processed by the Edge AI Swelling Prediction Engine (102), which utilizes a lightweight machine learning prediction model operating entirely on-device cloud connectivity. The prediction engine analyzes temperature rise rates, charging current deviations, voltage fluctuation patterns, workload intensity, battery aging indicators, and spatial micro thermal gradients (103) in order to generate a continuously updated Thermal Stress Score and Swelling Probability Score ranging from low-risk to high-risk operational states. The Swelling Probability Score is dynamically computed using weighted multi-parameter thermal analytics and represents the likelihood of battery swelling or abnormal thermal escalation occurring within a predicted future interval.

Simultaneously, the Micro Thermal Gradient Analysis module (103) operates in parallel with the prediction engine and detects non-uniform heat distribution patterns across consecutive sensor readings as early-stage indicators of internal electrochemical instability. This mechanism enables earlier warning capability compared to conventional systems that rely exclusively on absolute temperature thresholds.

The Behavioral Charging Learning module (104) continuously constructs and updates a personalized charging behavior profile by passively observing the user's charging habits and device usage patterns over time. The module records charging schedules, charging duration, State of Charge variation, active applications during charging, CPU and GPU load characteristics, and thermal response behavior across repeated charging sessions. Using this accumulated behavioral information, a lightweight on-device classifier predicts the expected thermal trajectory of upcoming charging sessions and enables pre-emptive charging optimization before dangerous thermal conditions arise. All behavioral profiles are securely stored locally on the device and are never transmitted externally.

The App-Level Heat Mapping module (104) maintains a continuously updated on-device thermal contribution database for installed applications. For each application executed during charging sessions, the system records temperature rise rate, CPU/GPU utilization contribution, peak thermal stress generation, and frequency of association with elevated Thermal Stress Scores. When a thermally intensive application becomes active during charging, the CPU/GPU-Aware Charging Control mechanism (105c) dynamically adjusts charging current (105a), charging rate, and workload distribution (105c) in proportion to the application's thermal risk profile in order to minimize thermal stacking caused by simultaneous computational heat generation and battery charging heat accumulation.

The Battery Age Compensation module (105) continuously recalibrates intervention thresholds based on cumulative charge cycle count, capacity fade percentage, historical thermal stress exposure, and long-term degradation trends. As the battery ages and its thermal tolerance decreases, the system proportionally strengthens protective interventions and reduces allowable charging stress under comparable operating conditions.

The Self-Learning Feedback Engine (109) employs a lightweight on-device reinforcement learning agent that continuously improves charging optimization decisions and prediction accuracy through operational feedback collected during normal device usage. Positive reward signals are generated when protective interventions successfully stabilize thermal conditions, while corrective feedback signals are generated when interventions are insufficient or unnecessarily conservative. Using accumulated feedback data, the reinforcement learning agent dynamically refines Thermal Stress Score weighting parameters, Swelling Probability Score thresholds, charging optimization policies, and behavioral adaptation models throughout the operational lifetime of the device.

The Adaptive Charging Orchestration Layer (105) may implement the charging interventions through Reduce Charging Current (105a), Pause/Resume Charging (105b), and CPU/GPU Workload Control (105c). The Reduce Charging Current function (106) adjusts the charging current in response to the determined thermal and swelling risk. The Pause/Resume Charging function (107) selectively pauses or resumes charging based on the determined risk state. The CPU/GPU Workload Control function (108) coordinates processor workload conditions with charging operation to reduce concurrent heat generation.

The Battery Safety Response Layer (110) is configured to execute the determined protective intervention based on the outputs of the Edge AI Analytics Layer (102), Micro Thermal Gradient Analysis Layer (103), Behavioral Intelligence & App Thermal Analysis Layer (104), Adaptive Charging Orchestration Layer (105), and Self-Learning Optimization Layer (109).

An Emergency Safety Override Layer (110) is additionally incorporated to ensure fail-safe operation under extreme thermal conditions. If hardware-level critical temperature or voltage limits are exceeded, the system immediately overrides AI based optimization logic and activates direct charging interruption and device protection protocols to prevent hazardous operating conditions.

Through the continuous interaction of predictive thermal analytics (102), behavioral charging intelligence (104), application-aware thermal orchestration (104), adaptive charging optimization (105), battery aging compensation (105), reinforcement learning-based self improvement (109), and fail-safe protection mechanisms (110), the proposed invention forms a closed-loop

autonomous battery safety orchestration framework operating entirely on-device without requiring additional hardware, internet connectivity, cloud processing, or user intervention. The invention thereby provides a more intelligent, personalized, adaptive, private, and proactive approach to battery swelling prevention and thermal safety management in modern consumer portable electronic devices.

Unlike conventional battery management systems that rely on fixed threshold-based protection, the proposed invention integrates Edge AI-based swelling prediction (102), behavioral charging learning (104), application-level thermal analytics (104), battery age compensation (105), adaptive charging orchestration (105), and reinforcement learning optimization (109) within a unified software only framework operating entirely on-device. This integrated architecture enables proactive battery swelling prevention before irreversible thermal degradation occurs.

The proposed invention introduces an AI-Based Predictive Thermal Monitoring and Adaptive Charging Optimization System for preventing battery swelling in consumer smartphones, laptops, and other portable electronic devices. The system operates as a software-only, on-device framework that continuously monitors battery and device operating parameters, including temperature, voltage, current, State of Charge (SoC), State of Health (SoH), processor utilization, and charging behavior. By leveraging Edge Artificial Intelligence, the invention predicts thermal stress conditions and battery swelling risk before critical damage occurs, thereby enabling proactive battery protection and improved operational safety. The invention integrates multiple intelligent modules including predictive thermal analytics, behavioral charging learning, application-level thermal analysis, adaptive charging control, battery age compensation, and self-learning optimization. These modules work together to generate Thermal Stress Scores and Swelling Probability Scores, identify thermal contributions from active applications, learn individual user charging patterns, and dynamically optimize charging behavior according to real-time operating conditions. The framework continuously adapts its protection strategy based on battery health, workload conditions, and historical usage patterns to minimize thermal degradation and extend battery lifespan. Unlike conventional battery management systems that rely primarily on fixed threshold-based reactive protection mechanisms, the proposed invention adopts a predictive and adaptive approach to battery safety management. The system performs all monitoring, analysis, prediction, and decision-making processes directly on the device without

requiring cloud connectivity, external servers, internet access, or additional hardware sensors. Through the integration of Edge AI, adaptive charging orchestration, and continuous learning capabilities, the invention provides an intelligent, autonomous, and scalable solution for battery swelling prevention and thermal risk management in modern portable electronic devices. System Architecture The proposed invention adopts a layered Edge AI-Based Predictive Thermal Risk Assessment and Adaptive Charging Orchestration Architecture for preventing lithium-ion battery swelling in smartphones, laptops, and portable electronic devices. The architecture consists of seven interconnected layers that continuously acquire operational data, predict thermal risks, optimize charging behavior, and improve decision-making through self- learning mechanisms. The overall framework operates entirely on-device without requiring cloud connectivity, external hardware, or user intervention.

The proposed invention introduces an AI-Based Predictive Thermal Monitoring and Adaptive Charging Optimization System (101–110) for preventing battery swelling in consumer smartphones, laptops, and other portable electronic devices. The system operates as a software-only, on-device framework that continuously monitors battery and device operating parameters, including temperature, voltage, current, State of Charge (SoC), State of Health (SoH), processor utilization, and charging behavior (101). By leveraging Edge Artificial Intelligence (102), the invention predicts thermal stress conditions and battery swelling risk before critical damage occurs, thereby enabling proactive battery protection and improved operational safety.

The invention integrates multiple intelligent modules including predictive thermal analytics (102), behavioral charging learning (104), application-level thermal analysis (104), adaptive charging control (105), battery age compensation (105), and self-learning optimization (109). These modules work together to generate Thermal Stress Scores and Swelling Probability Scores (102), identify thermal contributions from active applications (104), learn individual user charging patterns (104), and dynamically optimize charging behavior according to real-time operating conditions (105). The framework continuously adapts its protection strategy based on battery health, workload conditions, and historical usage patterns to minimize thermal degradation and extend battery lifespan.

Unlike conventional battery management systems that rely primarily on fixed threshold-based reactive protection mechanisms, the proposed invention adopts a predictive and adaptive approach

to battery safety management. The system performs all monitoring, analysis, prediction, and decision-making processes directly on the device without requiring cloud connectivity, external servers, internet access, or additional hardware sensors. Through the integration of Edge AI (102), adaptive charging orchestration (105), and continuous learning capabilities (109), the invention provides an intelligent, autonomous, and scalable solution for battery swelling prevention and thermal risk management in modern portable electronic devices.

System Architecture

The proposed invention adopts a layered Edge AI-Based Predictive Thermal Risk Assessment and Adaptive Charging Orchestration Architecture (101–110) for preventing lithium-ion battery swelling in smartphones, laptops, and portable electronic devices. The architecture consists of seven interconnected layers that continuously acquire operational data, predict thermal risks, optimize charging behavior, and improve decision-making through self-learning mechanisms. The overall framework operates entirely on-device without requiring cloud connectivity, external hardware, or user intervention.

Data Acquisition Layer (101)

The Data Acquisition Layer (101) continuously collects real-time operational parameters from built-in battery sensors and operating system interfaces. The acquired parameters include battery temperature, charging voltage, charging current, State of Charge (SoC), State of Health (SoH), charge cycle count, CPU utilization, GPU utilization, active application information, and ambient temperature where available. The collected data is timestamped and stored in a rolling time series buffer for further processing by the predictive analytics engine (102).

Output:

1. Battery Sensor Data
2. Device Usage Data
3. Charging Statistics
4. Thermal Parameters

Edge AI Analytics Layer (102)

The Edge AI Analytics Layer (102) performs predictive thermal analysis using a lightweight machine learning prediction model deployed entirely on-device. The layer analyzes thermal patterns, charging anomalies, workload intensity, and battery health indicators to calculate thermal risk.

Thermal Stress Score (TSS)

The Thermal Stress Score quantifies the instantaneous thermal stress experienced by the battery.

$$TSS = w_1\left(\frac{dT}{dt}\right) + w_2(V_{dev}) + w_3(I_{dev}) + w_4(Grad) + w_5\,(Load_{ratio}\,)$$

Where:

1. dT/dt = Temperature rise rate
2. V_{dev} = Voltage deviation
3. I_{dev} = Current deviation
4. Grad = Micro thermal gradient
5. $Load_{ratio}$ = CPU/GPU workload ratio
6. w1, w2, w3, w4, w5 = weighting coefficients

Swelling Probability Score (SPS)

The Swelling Probability Score estimates the probability of battery swelling occurring in the near future.

$$SPS = \frac{1}{1 + e^{-k(TSS - TSS_{threshold})}}$$

Where:

1. k = sensitivity coefficient

2. $TSS_{threshold}$ = thermal stress threshold

The SPS value ranges between 0 and 1 and is used as the primary decision variable for charging intervention.

Micro Thermal Gradient Analysis Layer (103)

This layer detects localized thermal abnormalities that may indicate early electrochemical instability before critical temperature levels are reached.

$$Grad = \frac{T_{max} - T_{min}}{T_{nominal}}$$

Where:

1. T_{max} = maximum observed temperature

2. T_{min} = minimum observed temperature

3. $T_{nominal}$ = nominal operating temperature

A higher gradient value indicates non-uniform heat distribution and increased thermal risk.

Behavioral Intelligence and Application Thermal Analysis Layer (104)

This layer continuously learns user charging behavior and identifies applications contributing to thermal stress. The Behavioral Learning Module analyzes:

1. Charging duration

2. Charging frequency

3. Overnight charging patterns

4. Gaming while charging

5. CPU/GPU workload trends

Simultaneously, the App-Level Heat Mapping Module computes an Application Thermal Contribution Score.

$$ATCS(\text{app}) = \frac{avg\left(\frac{dT}{dt} \mid app\ active\right)}{avg\left(\frac{dT}{dt} \mid all\ sessions\right)}$$

Adaptive Charging Orchestration Layer (105)

The Adaptive Charging Orchestration Layer (105) dynamically adjusts charging behavior based on thermal risk predictions, battery health conditions, and application thermal impact.

Battery Age Compensation Factor (ACF)

$$ACF = 1 + \alpha(1 - SoH) + \beta\left(\frac{Cycles}{Max_{Cycles}}\right)$$

The thermal threshold is recalibrated as:

$$\text{Adjusted}_{TSS_{threshold}} = \frac{\text{Base}_{TSS_{threshold}}}{ACF}$$

Where:

1. SoH = State of Health

2. Cycles = charge cycle count

3. MaxCycles= expected battery lifespan cycles

4. α, β = aging coefficients

Adaptive Charging Current (ACC)

$$ACC = I_{max} \times (1 - SPS) \times \left(\frac{1}{ACF}\right)$$

Where:

1. I_{max} = maximum charging current

The charging current is automatically reduced as swelling probability and battery aging increase.

The Application thermal contribution is also incorporated:

$$ACC_{adjusted} = ACC\left(\frac{1}{ATCS(app)}\right)$$

This prevents thermal stacking caused by simultaneous charging and computational workloads.

Self-Learning Optimization Layer (109)

The Self-Learning Optimization Layer (109) utilizes lightweight reinforcement learning to continuously improve charging optimization decisions.

Reinforcement Learning Reward Function

$$R = +1 \; if \; TSS_{after} < TSS_{before} \; and \; SPS_{after} < threshold$$

$$R = -0.5 \; if \; TSS_{after} > TSS_{before}$$

$$R = -0.2 \; if \; intervention \; was \; unnecessary$$

Weight updates are performed using:

$$w_{new} = w_{old} + learning_{rate} \times R \times gradient$$

This enables continuous refinement of thermal prediction accuracy and charging decisions.

Battery Safety Response Layer (110):

The Battery Safety Response Layer (110) executes preventive and emergency actions based on outputs generated by previous layers. The layer performs:

1. Charging current reduction

2. Charging pause and resume control

3. CPU/GPU workload coordination

4. Thermal risk notifications

5. Swelling prevention interventions

6. Emergency charging shutdown under critical conditions Additionally, charger stability is monitored using the Charger Quality Detection metric.

Charger Quality Detection (CQD)

$$CQD = \frac{std_deviation(Voltage_{readings})}{V_{nominal}}$$

Abnormal CQD values indicate unstable charging conditions and trigger additional protection mechanisms.

EXPERIMENTAL RESULTS

The proposed BatteryThermalAI framework was experimentally validated using a Windows-based laptop operating under real-world conditions. During the testing period, the device remained in an idle state and operated on battery power. The Battery State of Charge (SoC) was maintained at 100%, while the State of Health (SoH) remained at 100.0%, indicating an excellent battery condition with no observable degradation effects.

The recorded battery temperature remained highly stable throughout the monitoring period, varying between 31.94°C and 32.42°C. Such minimal temperature variation demonstrates stable thermal behavior and confirms that the system can continuously monitor battery conditions without introducing significant computational overhead.

The Thermal Stress Score (TSS) ranged from 0.1497 to 0.1516 and consistently remained within the low-risk region. These values indicate that the combined influence of temperature rise rate, voltage deviation, current deviation, micro thermal gradient, and processor workload remained significantly below critical operating thresholds. The low TSS values validate the effectiveness of the proposed thermal stress assessment model in accurately characterizing battery operating conditions. The Swelling Probability Score (SPS) was observed between 3.91% and 3.97%, indicating an extremely low likelihood of battery swelling under the tested conditions. The consistently low SPS values confirm that the predictive swelling assessment model successfully translates thermal stress measurements into meaningful risk estimates while maintaining high sensitivity to potential thermal abnormalities.

The Adaptive Charging Current (ACC) was dynamically computed between 2.881 A and 2.883 A., corresponding to a charging current optimization of approximately 97% of the maximum allowable charging current. This demonstrates that the proposed adaptive charging mechanism can

continuously generate safe charging recommendations based on real-time battery conditions. Even though the laptop was not actively charging during the experiment, the framework successfully performed charging current optimization calculations in real time.

The Micro Thermal Gradient value remained approximately 0.0491, indicating a highly uniform thermal distribution across the monitored battery region. This suggests the absence of localized thermal hotspots and confirms stable electrochemical behavior within the battery.

Additionally, the Battery Age Compensation Factor (ACF) remained at 1.0 due to the battery's excellent health condition, indicating that no age-related threshold adjustments were required. The App Thermal Contribution Score (ATCS) was recorded as 1.0, reflecting negligible thermal impact from the active application, which was identified as the system idle process.

Furthermore, the Charger Quality Detection (CQD) value remained approximately 0.00146, indicating stable voltage conditions and the absence of significant charger-related anomalies. The live trend analysis of temperature, Thermal Stress Score, and Swelling Probability Score exhibited nearly flat profiles throughout the testing period, demonstrating stable system operation and low processing overhead suitable for continuous Edge AI deployment.

Overall, the experimental results validate the practical feasibility and effectiveness of the proposed BatteryThermalAI framework. The prototype successfully performed real-time thermal monitoring, thermal stress assessment, swelling probability prediction, adaptive charging current computation, battery age compensation, application-level thermal analysis, and charger quality evaluation using only on-device resources. The consistently low TSS and SPS values demonstrate the capability of the proposed system to provide proactive battery safety management and battery swelling prevention while maintaining efficient operation on consumer laptops.

CLAIMS

We claim,

1. A software-implemented, on-device artificial intelligence (AI) based predictive thermal monitoring and adaptive charging optimization system for preventing swelling of a rechargeable battery of a portable electronic device, the system comprising:

a) a Data Acquisition Layer (101) configured to continuously acquire and timestamp battery and device operating parameters comprising battery temperature, voltage, charging current, State of Charge (SoC), State of Health (SoH), charge cycle count, CPU utilization, GPU utilization, active application information, charging duration, and ambient temperature;

b) an Edge AI Analytics Layer (102) configured to process the acquired parameters using an on-device machine learning model and determine a Thermal Stress Score (TSS) and a Swelling Probability Score (SPS) indicative of a predicted likelihood of thermal instability or battery swelling within a future time interval;

c) a Micro Thermal Gradient Analysis Layer (103) configured to identify localized and non-uniform thermal variations from temporally successive thermal measurements and provide an indication of an early-stage thermal instability;

d) a Behavioral Intelligence and App Thermal Analysis Layer (104) configured to learn user-specific charging behavior and device usage patterns and determine a thermal contribution associated with one or more active applications based on at least application activity, processor utilization, temperature rise, and historical thermal behavior;

e) an Adaptive Charging Orchestration Layer (105) configured to determine a charging intervention based on at least the TSS, the SPS, the learned charging behavior, application thermal contribution, processor workload, and battery condition, wherein the charging intervention comprises selectively reducing charging current (105a), pausing or resuming charging (105b), and controlling CPU/GPU workload (105c);

f) a Battery Age Compensation mechanism (105) configured to modify one or more thermal-risk thresholds, SPS thresholds, and charging intervention parameters according to battery

aging indicators comprising charge cycle count, capacity degradation, historical thermal stress exposure, or degradation trends;

g) a Self-Learning Optimization Layer (109) configured to update prediction parameters and charging optimization policies based on operational feedback associated with execution of the charging intervention; and

h) a Battery Safety Response Layer (110) configured to execute the determined charging intervention and provide a safety response according to the predicted thermal or swelling risk,

i) wherein the system continuously performs acquisition, prediction, risk assessment, charging optimization, intervention, and feedback in a closed-loop manner entirely on the portable electronic device, thereby predicting and mitigating battery swelling before occurrence of a critical thermal condition without requiring cloud processing, internet connectivity, or additional external sensing hardware.

2. The system as claimed in claim 1, wherein the Data Acquisition Layer (101) maintains a rolling time-series buffer containing timestamped battery, thermal, charging, processor, application, and usage parameters for determining temporal changes and trends associated with battery thermal behavior.

3. The system as claimed in claim 1, wherein the Edge AI Analytics Layer (102) determines the TSS and SPS using weighted multi-parameter analysis comprising at least temperature rise rate, charging current, voltage fluctuation, workload intensity, battery aging information, and thermal gradient information, and classifies the battery into a plurality of progressively increasing thermal-risk states.

4. The system as claimed in claim 1, wherein the Micro Thermal Gradient Analysis Layer (103) compares thermal measurements obtained at different time instances to identify a localized temperature gradient, a non-uniform heat distribution, or an abnormal rate of localized temperature change preceding an increase in overall battery temperature.

5. The system as claimed in claim 1, wherein the Behavioral Intelligence and App Thermal Analysis Layer (104) generates a personalized charging behavior profile based on at least charging

schedules, charging duration, SoC variation, applications executed during charging, CPU/GPU utilization, and thermal responses associated with previous charging sessions, and uses the personalized charging behavior profile to predict a thermal trajectory of a subsequent charging session.

6. The system as claimed in claim 1, wherein the Behavioral Intelligence and App Thermal Analysis Layer (104) maintains an application-level thermal contribution database identifying, for respective applications, at least temperature rise rate, CPU/GPU utilization contribution, peak thermal stress, and historical association with elevated TSS or SPS values, and wherein the Adaptive Charging Orchestration Layer (105) modifies charging operation in response to the identified application-level thermal contribution.

7. The system as claimed in claim 1, wherein the Adaptive Charging Orchestration Layer (105) coordinates charging operation with processor workload by selectively executing the Reduce Charging Current function (105a), the Pause/Resume Charging function (105b), and the CPU/GPU Workload Control function (105c), wherein the CPU/GPU Workload Control function (108) modifies processor workload conditions to reduce simultaneous computational heat generation and charging-related heat generation.

8. The system as claimed in claim 1, wherein the Battery Age Compensation mechanism (105) dynamically recalibrates the charging intervention according to a combination of charge cycle count, capacity fade, historical thermal exposure, and battery degradation trend, such that an allowable charging stress is progressively reduced as battery aging increases.

9. The system as claimed in claim 1, wherein the Self-Learning Optimization Layer (109) comprises a lightweight on-device reinforcement learning agent configured to generate a positive reward in response to an intervention that stabilizes a thermal condition and a corrective feedback signal in response to an intervention that is insufficient or unnecessarily conservative, and to update at least TSS weighting parameters, SPS thresholds, charging policies, or behavioral adaptation parameters based on the feedback.

10. The system as claimed in claim 1, further comprising an Emergency Safety Override Layer (110) configured to override the AI-based charging optimization when a hardware-level critical temperature or voltage limit is exceeded and to directly initiate charging interruption and/or a

device protection protocol, thereby providing a fail-safe response independently of the predictive AI decision.

AI-Based Predictive Thermal Monitoring and Adaptive Charging Optimization System for Preventing Battery Swelling in Consumer Smartphones and Laptops

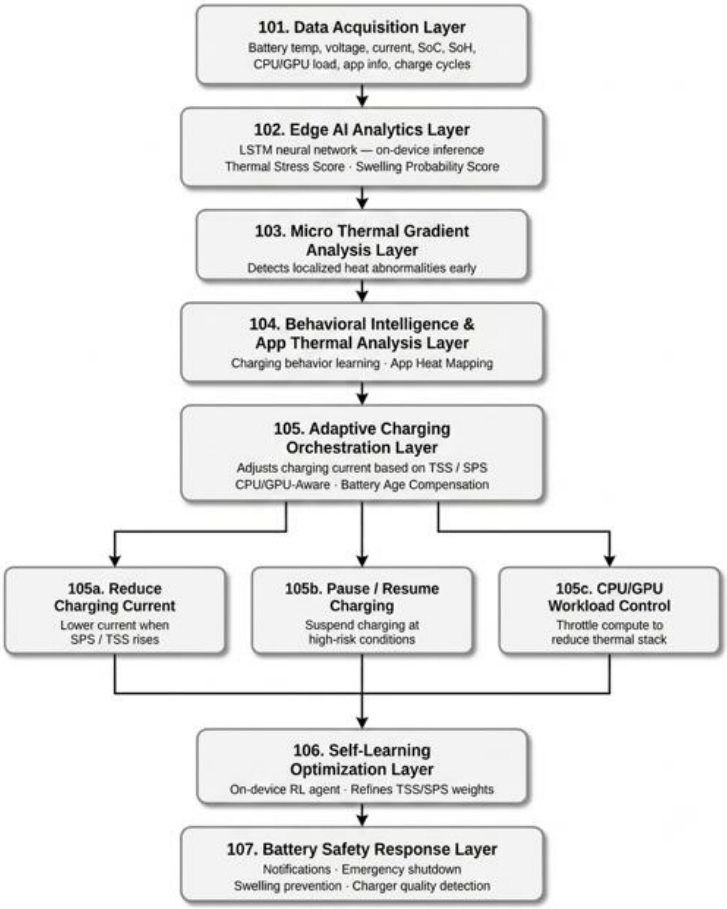
ABSTRACT

The present invention provides an AI-Based Predictive Thermal Monitoring and Adaptive Charging Optimization System for preventing battery swelling in smartphones, laptops, and portable electronic devices. The system operates as a software-only, on-device Edge AI framework that continuously acquires battery temperature, voltage, current, State of Charge (SoC), State of Health (SoH), processor workload, application activity, and charging behavior. A predictive analytics layer generates a Thermal Stress Score (TSS) and Swelling Probability Score (SPS) to identify potential thermal instability and swelling risk before critical conditions arise. The system further incorporates behavioral charging learning, application-level heat mapping, CPU/GPU-aware charging control, battery age compensation, micro thermal gradient analysis, adaptive charging orchestration, and self-learning optimization. Based on the predicted risk, charging current, charging rate, and charging operation are dynamically controlled. The framework operates offline without cloud connectivity, external hardware, or internet access, thereby providing proactive, personalized, autonomous, and privacy-preserving battery safety management.

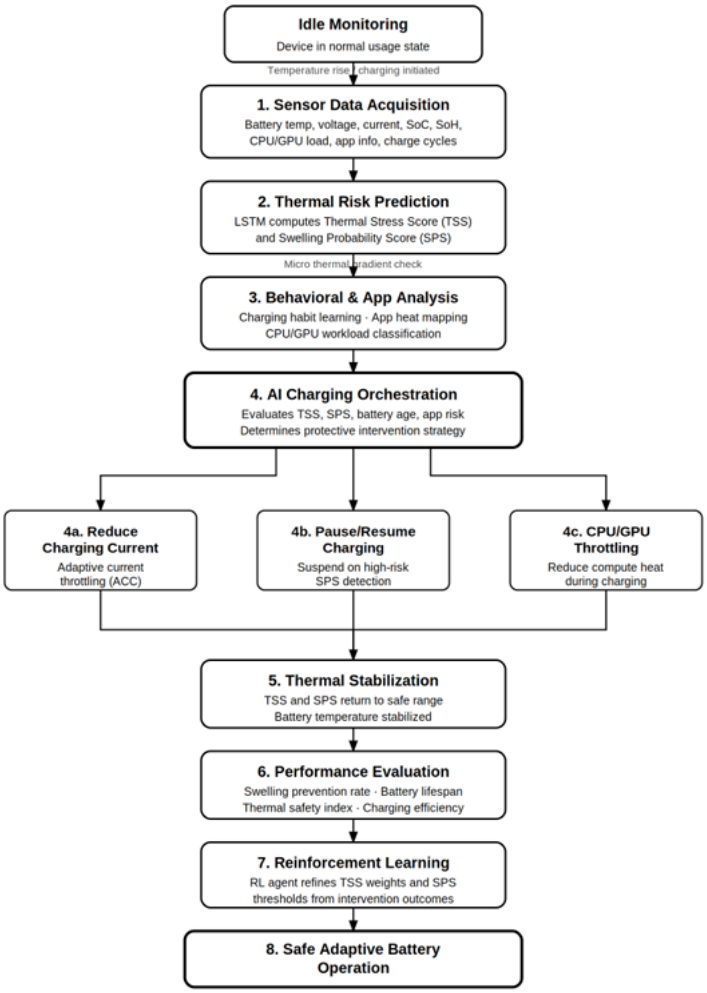
Date: Signature of Applicant

600 **DRAWING**

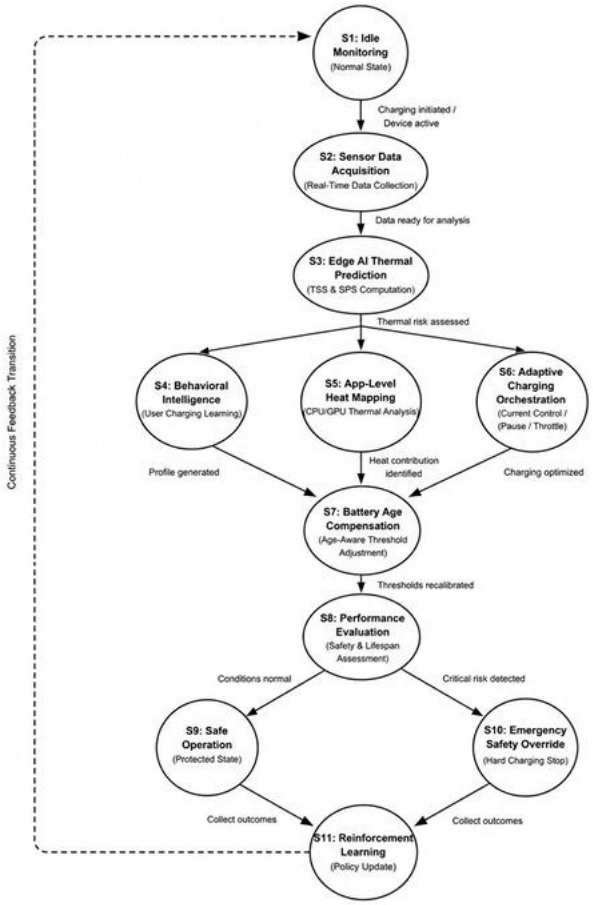
Fig. 1



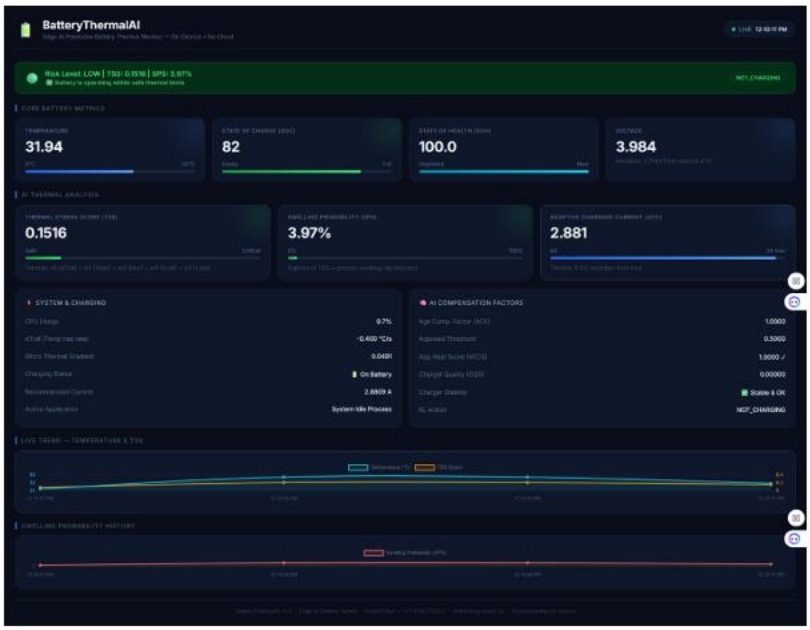
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Idle Monitoring
Device in normal usage state
Temperature rise / charging initiated
1. Sensor Data Acquisition
Battery temp, voltage, current, SoC, SoH,
CPU/GPU load, app info, charge cycles
2. Thermal Risk Prediction
LSTM computes Thermal Stress Score (TSS)
and Swelling Probability Score (SPS)
Micro thermal gradient check
3. Behavioral & App Analysis
Charging habit learning · App heat mapping
CPU/GPU workload classification
4. AI Charging Orchestration
Evaluates TSS, SPS, battery age, app risk
Determines protective intervention strategy
4a. Reduce Charging Current
Adaptive current throttling (ACC)
4b. Pause/Resume Charging
Suspend on high-risk SPS detection
4c. CPU/GPU Throttling
Reduce compute heat during charging
5. Thermal Stabilization
TSS and SPS return to safe range
Battery temperature stabilized
6. Performance Evaluation
Swelling prevention rate · Battery lifespan
Thermal safety index · Charging efficiency
7. Reinforcement Learning
RL agent refines TSS weights and SPS
thresholds from intervention outcomes
8. Safe Adaptive Battery Operation

610 **Fig. 3** **Page 3 of 4**



615



620

Metric	Value	Status
Temperature	32.42 °C	✓
Voltage	3.684 V	
SoC	57 %	
SoH	100.0 %	
dT/dt	-0.27 °C/s	
CPU Usage	12.1 %	
Thermal Stress (TSS)	0.1497	LOW
Swelling Prob (SPS)	0.0391	LOW
Age Factor (ACF)	1.0	
ATCS (app heat)	1.0	✓
Rec. Current (ACC)	2.8827 A	
Charger Quality	CQD=0.00146	✓ OK
Active App	System Idle Process	
Action	NOT_CHARGING	